Hedging Uniswap V4 Liquidity Positions with GMX Perpetuals

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Abstract

This document examines the use of perpetual futures to hedge Uniswap V4 concentrated liquidity positions (LP). By combining short perpetual strategies and Xtreamly volatility prediction, liquidity providers can reduce impermanent loss and even gain profits on price fluctuations. The impact of Xtreamly volatility predictions, perpetuals leverage, funding costs, and liquidation risks is discussed for effective risk management. Finally we apply hedging to historical LP positions and present results of possible overall improvement.

1 Hedging Component

We propose a hedging mechanism of Uniswap LP positions with GMX perpetuals. We state, that by combining these 2 instruments together we can reduce the risk, and even improve the overall performance of the position.

1.1 Uniswap LP Position

1.1.1 Concept

Consider a Uniswap V4 LP position that is providing liquidity between the lower price bound and upper bound. Without loss of abstraction, suppose we are providing liquidity in the ETH/USDT pool:

- **ETH (token0)**: The volatile asset.
- **USDT (token1)**: The stable asset.

1.1.2 Collected fees

These are income of LP position as a reward function for applying tokens and, add up to final profitability of LP Position. For the sake of simplicity on PnL we will skip this variable here as the main focus remains on impermanent loss.

1.1.3 Uniswap fees

As the LP payoff depends on which region the price P falls into we shall skip Uniswap and Gas fees for entering LP position.

1.1.4 Impermanent Loss

Impermanent loss (IL) is when the prices of tokens in a liquidity pool change from what they were when the liquidity was added, resulting in a loss of value in the LP position.

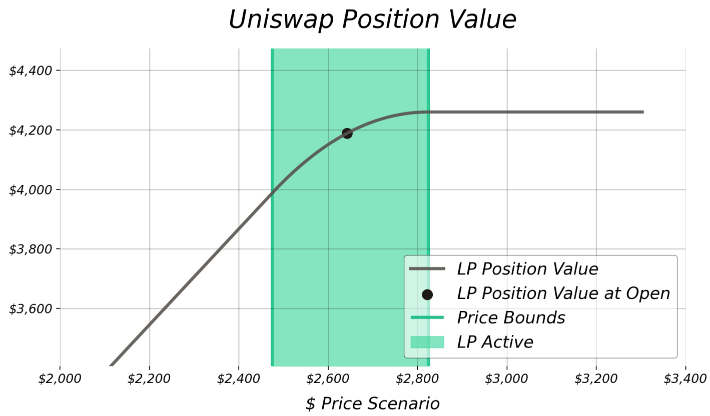

Figure 1: The LP position value for P price scenarios.

Depending on the final P price, it is clear that for downgrading price, the LP Value is degrading, resulting in impermanent loss. This phenomenon creates a demand for portfolio construction that would balance off the price, creating more market neutral position, and enable holder of LP to define profitability through collected fees.

1.2 GMX Perpetual Futures for Hedging

To hedge the ETH/USDT Uniswap V4 LP position, we propose using a perpetual futures contract on the GMX platform. Perpetual futures are derivative contracts that allow traders to speculate on the price of an asset (in this case, ETH) without an expiration date, making them a flexible tool for hedging. Unlike traditional futures, perpetuals are designed to closely track the spot price of the underlying asset through a funding rate mechanism, which balances the positions of long and short traders. In the context of hedging a Uniswap LP position, a short perpetual futures position can help mitigate the impermanent loss caused by ETH price fluctuations.

Let's define the fundamental components of a GMX perpetual futures contract in this hedging strategy:

- **Entry Price** (P_0): The price at the time opening. This serves as the reference price for calculating the perpetual PnL.
- **Mark Price** (P): The current price derived from an oracle.
- **Collateral** (M): The funding (in USDT) posted to open and maintain the perpetual position. The margin determines the leverage of the position and the risk of liquidation.
- **Leverage** (L): The ratio of the notional value of the position to the margin posted, i.e., Leverage $= \frac{\alpha P_0}{M}$. Higher leverage amplifies both potential gains and losses, increasing the risk of liquidation.
- **Notional Size** (α): The size of the perpetual futures position as a combination of leverage and collateral $\alpha = L \cdot M$
- **Liquidation Price** (P_{liq}): The price of ETH at which perpetuals goes in the liquidation of the position. On GMX,

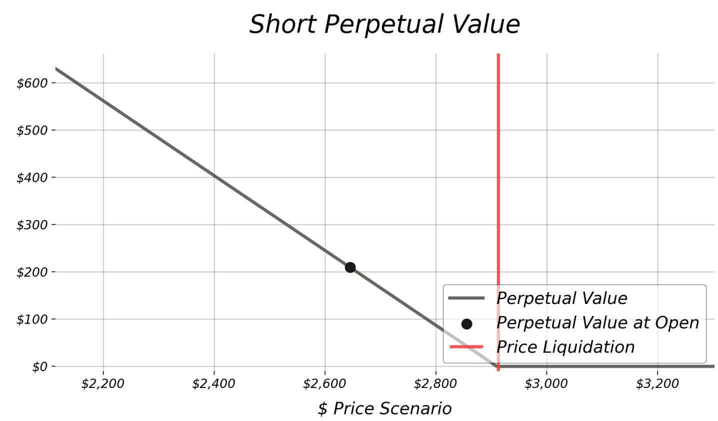

Figure 2: The Perpetual position value for P price scenarios.

the liquidation price depends on the leverage, margin, and fees.[1]

- **Funding Rates**: A periodic payment (charged every 1 hour on GMX) to keep the perpetual futures price aligned with the spot price (can be positive or negative).

1.2.1 Short Perpetual

By taking a short perpetual position on ETH, the LP can offset losses from impermanent loss when the ETH price increases. For example, if the ETH price rises, the LP position suffers impermanent loss due to the rebalancing of ETH and USDT in the pool, but the short perpetual position gains in value, compensating for this loss. However, this strategy introduces risks such as liquidation (if the ETH price rises too far) and funding costs (if the funding rate favors longs).

1.3 Xtreamly Forecast

1.3.1 Concept

To avoid liquidation we use Xtreamly volatility prediction to define healthy levels of leverage in order for the position not to get liquidated before protecting against IL. This will be discussed further in details on back testing strategy.

1.3.2 Market Status

As part of Xtreamly forecasting ability, is to define market status at each moment in time into 3 categories:

- **Low Volatility**: When large price movement is not expected.
- **Medium Volatility**: When moderate price movement is not expected.
- **High Volatility**: When large price movement are expected.

[1] In practice, exchanges like GMX may implement partial liquidations, maintenance margins, or additional fees to mitigate risk, but for simplicity, we assume that once the ETH price P reaches P_{liq}, the entire margin M is lost.

1.3.3 Dynamic Leverage

While Xtreamly cannot predict future price movement to the extend of significant precision, Xtreamly can use its volatility information to define leverage levels depending on price movement. This way, Xtreamly can avoid pre early liquidations of price reaching liquidation before closing of perpetual by dynamically defining leverage levels at opening perpetuals. This way the hedged position, whenever volatility is to predicted to be low, the higher leverage makes hedged position to be more profitable.

1.4 Combining LP and Perpetual

In this section, we analyze the combination of a Uniswap V4 concentrated liquidity (CL) position with a perpetual hedge. The goal is to dynamically hedge the price exposure of the LP position while maintaining capital efficiency.

1.4.1 Liquidation Risks

However, this strategy is not without risks. If the price oscillates too frequently around the entry levels of the perpetual positions, liquidations can occur before meaningful profits are realized. This would result in accumulated losses, as the hedges fail to serve their intended purpose of long-term exposure management. Xtreamly volatility prediction is not considered as we focus on PnL of hedged position and best proportions of the hedging. Therefore, we shall skip theses risks in this subsection.

Thus, to maximize the effectiveness of this combined approach, it is crucial to:

- Optimize hedge sizing to avoid excessive liquidation risk while maintaining capital efficiency.
- Adjust leverage cautiously, as higher leverage increases sensitivity to liquidation but also amplifies returns when successful.
- Backtest under different market conditions to analyze whether the perpetual hedges improve overall strategy performance or lead to excessive churn.

Overall, the combined approach offers significant downside protection and potential for additional upside capture, but requires careful risk management to prevent liquidation losses from overwhelming the benefits of the hedging strategy.

1.4.2 Considerations for Hedge Optimization

The effectiveness of this strategy depends on proper hedge allocation and leverage management:

- **Collateral allocation:** The fraction of total capital allocated to the hedge determines its impact on the overall payoff.
- **Leverage choice:** Higher leverage increases capital efficiency but also raises funding costs and liquidation risk.
- **Funding rate impact:** Perpetual futures often have non-zero funding rates, which can accumulate over time, affecting profitability.
- **Backtesting necessity:** The interaction between CL payoffs and perpetual hedges needs to be tested across various market conditions to ensure robustness.

By systematically balancing these parameters, liquidity providers can dynamically adjust their hedge exposure, optimizing for capital efficiency while managing risk exposure.

2 Hedging Mathematics

2.1 Uniswap Position

Let:

- P_L: The lower bound price of the LP position (USDT per ETH).
- P_U: The upper bound price of the LP position (USDT per ETH).
- P: The current market price of ETH in USDT.
- L: The *constant* liquidity associated with the position.
- X_0: The amount of ETH (token0) held at the time the position is set, related to P_U and P_L.
- Y_0: The amount of USDT (token1) held at the time the position is set, also related to P_U and P_L.

2.1.1 Three Cases for the LP Value

Case 1: Within Range $P_L \leq P \leq P_U$

When the market price P lies within the liquidity provider's chosen range $[P_L, P_U]$, the position consists of a mix of ETH and USDT. The exact amounts of each asset are determined by Uniswap's liquidity formula.

At any price P in this range:

$$X(P) = L\left(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{P_U}}\right), \qquad Y(P) = L\left(\sqrt{P} - \sqrt{P_L}\right).$$

where:

- $X(P)$ represents the amount of ETH (token0) held at price P.
- $Y(P)$ represents the amount of USDT (token1) held at price P.
- L is the total liquidity provided to the range.

The total value of the LP position at price P is then:

$$V(P) = X(P) \cdot P \; + \; Y(P).$$

Expanding:

$$V(P) = L\left(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{P_U}}\right)P + L\left(\sqrt{P} - \sqrt{P_L}\right), \quad \text{for } P_L \leq P$$

Case 2: Below Range $P < P_L$.

When the price drops below P_L, the liquidity position has been fully converted into ETH (token0). This happens because as price decreases, the Uniswap AMM mechanism sells all available USDT in exchange for ETH.

The ETH amount is locked at the value it had at P_L, which is:

$$X(P_L) = L\left(\frac{1}{\sqrt{P_L}} - \frac{1}{\sqrt{P_U}}\right).$$

Since the LP now holds only ETH, the total position value in USD is simply the ETH amount multiplied by the current price P:

$$V(P) = X(P_L) \cdot P.$$

Expanding:

$$V(P) = L\left(\frac{1}{\sqrt{P_L}} - \frac{1}{\sqrt{P_U}}\right)P, \quad \text{for } P < P_L.$$

Case 3: Above Range $P > P_U$.

When the price moves above P_U, the liquidity position has been fully converted into USDT (token1). This happens because as price increases, the Uniswap AMM mechanism sells all available ETH in exchange for USDT.

The USDT amount is locked at the value it had at P_U, which is:

$$Y(P_U) = L\left(\sqrt{P_U} - \sqrt{P_L}\right).$$

Since the LP now holds only USDT, the total position value remains constant:

$$V(P) = Y(P_U).$$

Expanding:

$$V(P) = L\big(\sqrt{P_U} - \sqrt{P_L}\big), \quad \text{for } P > P_U.$$

Final Piecewise Formula. Bringing everything together, the complete value function for the LP position is:

$$V(P) = \begin{cases} L\left(\frac{1}{\sqrt{P_L}} - \frac{1}{\sqrt{P_U}}\right)P, & P < P_L, \\ L\Big(\big(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{P_U}}\big)P + \big(\sqrt{P} - \sqrt{P_L}\big)\Big), & P_L \le P \le P_U, \\ L\big(\sqrt{P_U} - \sqrt{P_L}\big), & P > P_U. \end{cases}$$

2.2 GMX Perpetual

We consider a perpetual futures contract to hedge ETH/USDT. Let:

- M: The margin (in USDT) posted for the perpetual position.
- α: The *notional* size of the perp (in ETH units).
- P_0: The entry price (USDT per ETH).
- P_{liq}: The *liquidation price*, where margin is depleted and lost.[2]

Short Perpetual. Suppose we *short* at price P_0. The PnL from the short is $\alpha\,(P_0 - P)$. We post margin M to cover losses if P rises. We define P_{liq} so that when $P = P_{\text{liq}}$, our margin is fully exhausted:

$$M + \alpha\,(P_0 - P_{\text{liq}}) = 0 \quad \Longrightarrow \quad P_{\text{liq}} = P_0 + \frac{M}{\alpha}.$$

Hence, our **short** payoff is

$$V_{\text{short}}(P) = \begin{cases} M + \alpha\,(P_0 - P), & P < P_{\text{liq}}, \\ -M, & P \ge P_{\text{liq}}. \end{cases}$$

Long Perpetual. If we *long* at P_0, the PnL is $\alpha\,(P - P_0)$. The margin M is depleted if P falls by enough. Setting $P = P_{\text{liq}}$ where margin is exhausted:

$$M + \alpha\,(P_{\text{liq}} - P_0) = 0 \quad \Longrightarrow \quad P_{\text{liq}} = P_0 - \frac{M}{\alpha}.$$

Hence, our **long** payoff is

$$V_{\text{long}}(P) = \begin{cases} -M, & P \le P_{\text{liq}}, \\ \alpha\,(P - P_0), & P > P_{\text{liq}}. \end{cases}$$

Interpretation. At liquidation ($P = P_{\text{liq}}$), the payoff *jumps* from some positive level (on the linear portion) down to $-M$, reflecting full loss of the margin. This discontinuous drop represents the complete loss of collateral at the liquidation boundary.

[2]In practice, exchanges might have partial liquidations or maintenance margins, but here we simplify: once P hits P_{liq}, the entire margin M is lost.

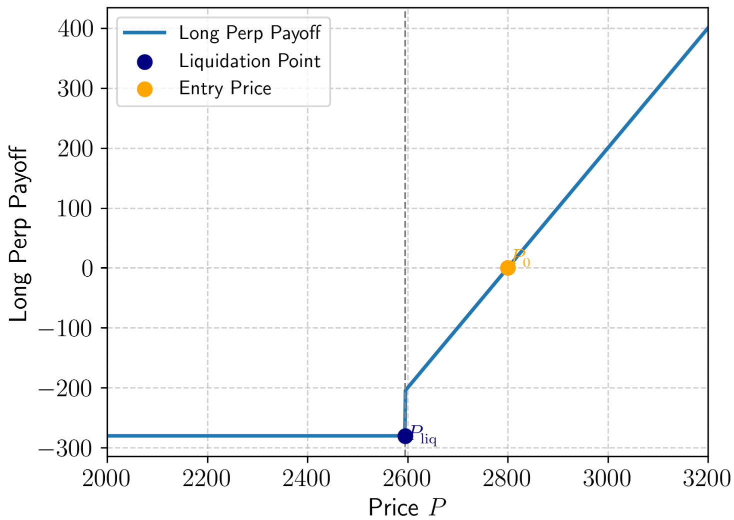

Figure 5: Piecewise payoff for the Long Perpetual, showing the drop to $-M$ at liquidation.

2.3 Plots on Combining LP and different Perpetuals

2.4 Long Perpetual Hedge with Concentrated Liquidity

In Figure 6, we illustrate the combined payoff of a long perpetual hedge with a Uniswap CL position. This approach is used when the AI expects volatility and wants to ensure exposure to price appreciation of the underlying asset (ETH in our example). Without a hedge, once the price moves above P_U, the LP position fully converts into USDT, leading to missed upside. By incorporating a long perpetual hedge, the strategy ensures that additional gains can be captured.

Key observations from Figure 6:

- The Uniswap CL position alone (dashed gray line) exhibits a flat payoff above P_U, since all liquidity has been converted to USDT.
- The long perpetual hedge introduces a positive slope above the entry price, capturing further upside.
- If the price declines and liquidation occurs, the strategy incurs a sharp loss at P_{liq}.

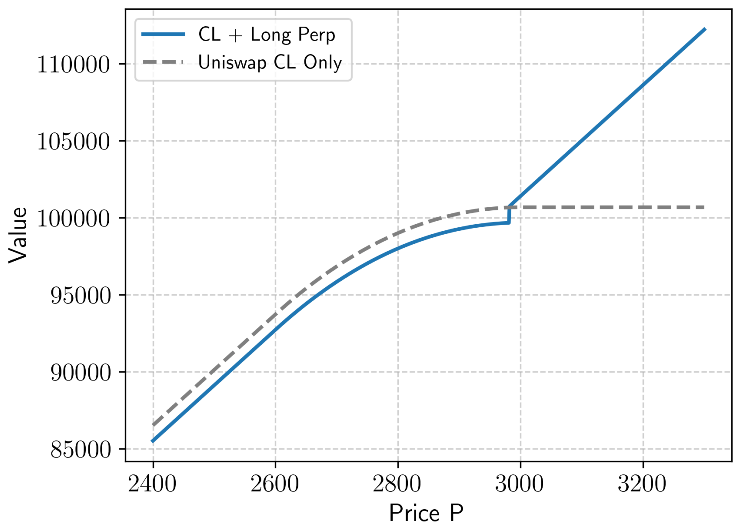

Figure 6: Example combined payoff: Concentrated liquidity plus a Long Perp.

2.5 Short Perpetual Hedge with Concentrated Liquidity

In Figure 8, we show the combined payoff of a short perpetual hedge with a Uniswap CL position. This strategy is employed when we seek to protect against downside risk. Without a hedge, if the price drops below P_L, the position converts entirely into ETH, exposing the LP to further losses. A short perpetual hedge compensates for these losses.

Key observations from Figure 8:

- The Uniswap CL position alone (dashed gray line) exhibits a linear decline below P_L due to full conversion into ETH.
- The short perpetual hedge introduces a slope below the entry price, mitigating further downside.
- If the price increases and liquidation occurs, the hedge incurs a sharp loss at P_{liq}.

Figure 7 illustrates the combined payoff when using both a short and long perpetual hedge alongside a Uniswap CL position. This configuration aims to provide protection on both the downside and upside, mitigating the risk of price movements that push the LP position out of range.

Key insights:

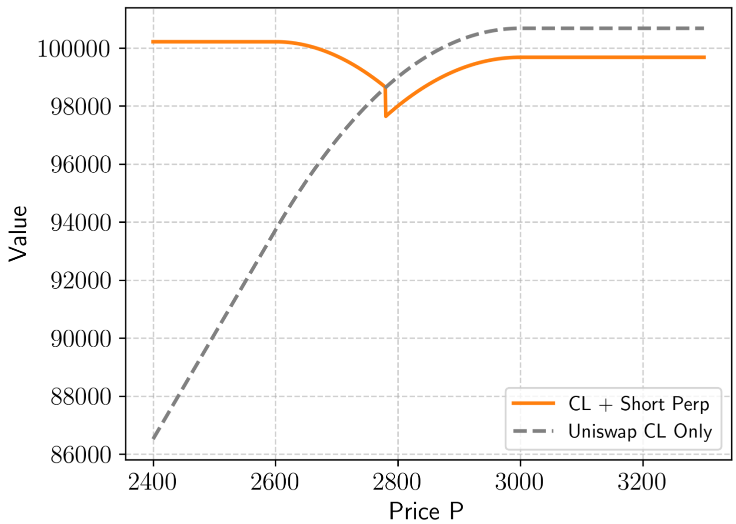

Figure 7: Example combined payoff: Concentrated liquidity plus a Short Perp.

- The long perpetual hedge compensates for missed upside when the price surpasses P_U, ensuring the strategy does not forgo further gains once the LP position is fully converted to USDT.
- The short perpetual hedge offsets losses when the price declines below P_L, counteracting the exposure to ETH that would otherwise be unhedged.
- The combined strategy flattens the risks associated with concentrated liquidity by maintaining a more balanced exposure across price ranges.

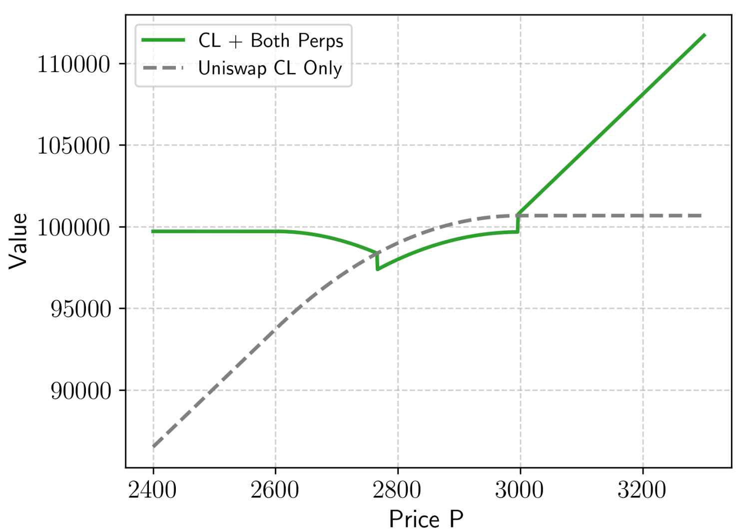

Figure 8: Example combined payoff: Concentrated liquidity plus a Short and Long Perp.